EMRECAN GULAY
Postdoctoral Researcher and Doctor of Arts

Human AI interaction is the broader research focus

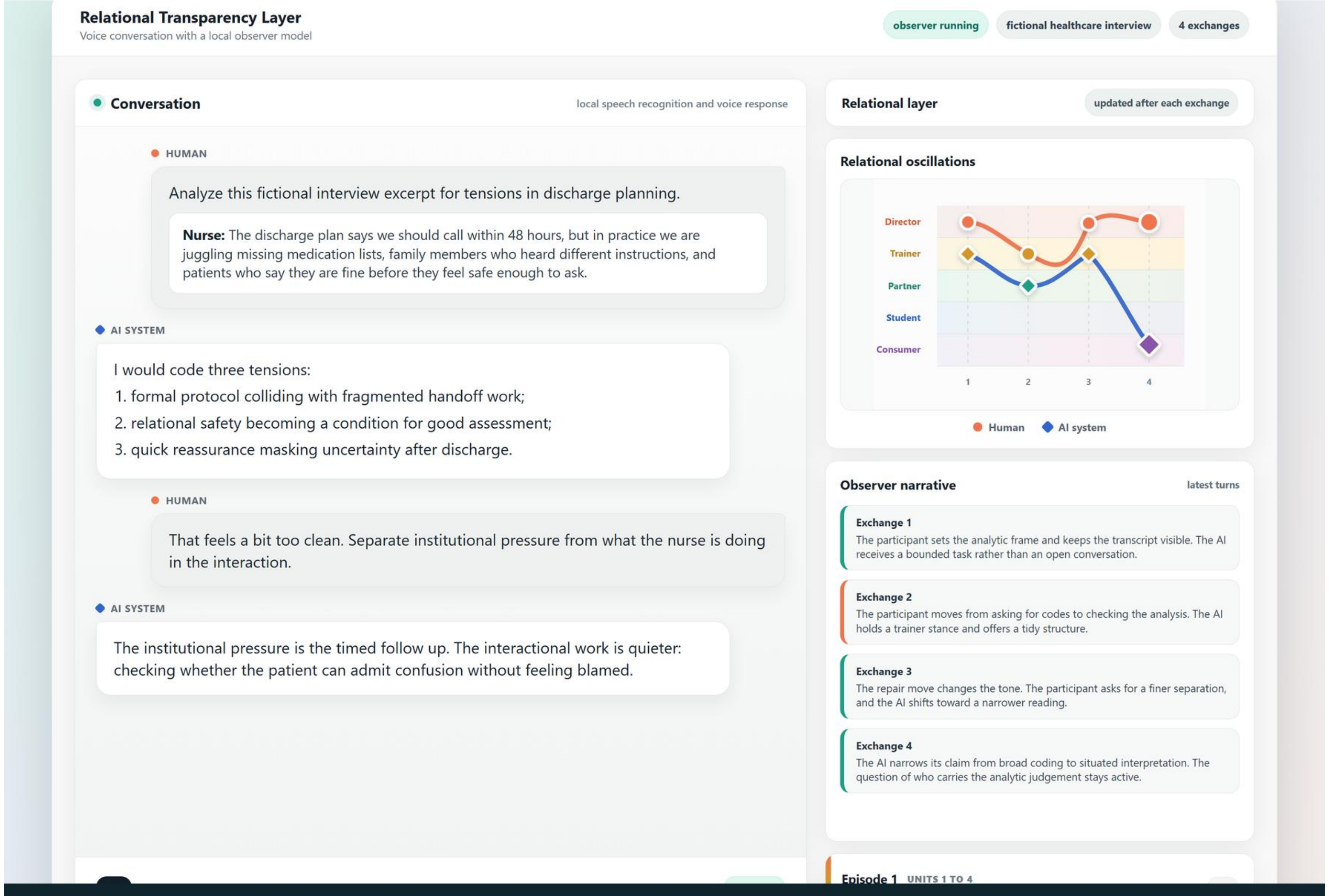
I study human AI interaction through qualitative inquiry, interaction evidence, and research prototypes.

CURRENT ROLE
Responsible Use of AI, Aalto University

RESEARCH LEADERSHIP
Project lead, Relational Transparency Layer

PUBLICATION STATUS
Accepted demo, NordiCHI 2026

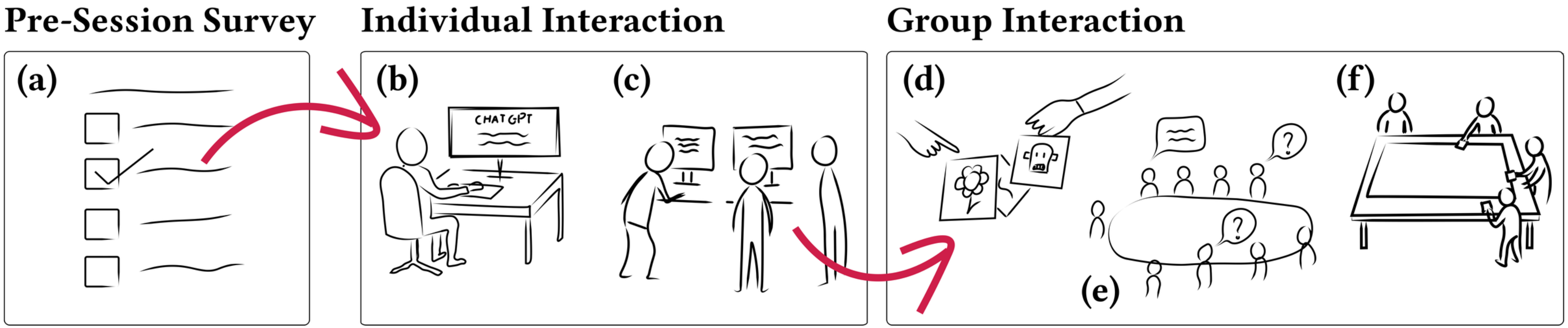
live portfolio | email



PARTICIPANT STUDY ¤ WORKING PROTOTYPE

01A Relational Dissonance
Participant research in AI assisted knowledge work

How do researchers describe AI, and how does that relation appear in their interaction?



STUDY

3 workshops, 22 qualitative researchers, and 4 AI systems.

Pre session survey, individual AI interaction, and group discussion.

EVIDENCE

Surveys, chat logs, image choices, group worksheets, and observations.

The quantitative measures provide descriptive context for qualitative interpretation.

ANALYSIS AND ROLE

Individual level interpretative phenomenological analysis and social level thematic analysis were integrated by the team.

First author in a four person interdisciplinary research team.

ANALYTIC SEQUENCE

01

Stated position

Pre session survey responses and participant selected images.

02

Enacted relation

Chat logs and observations show how roles shift during the task.

03

Integrated account

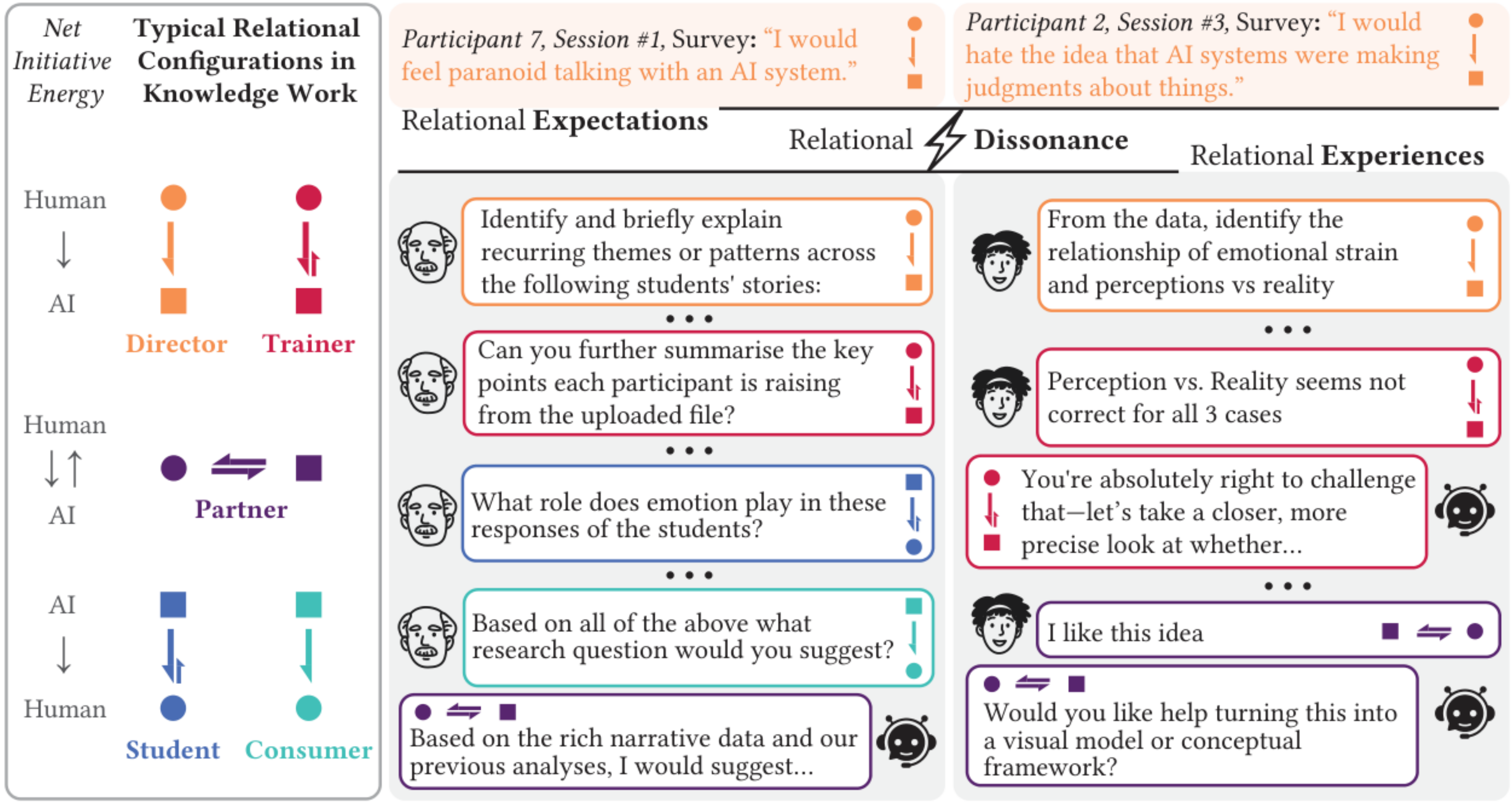
Individual and social analyses are brought together in team synthesis.

UNIT OF ANALYSIS

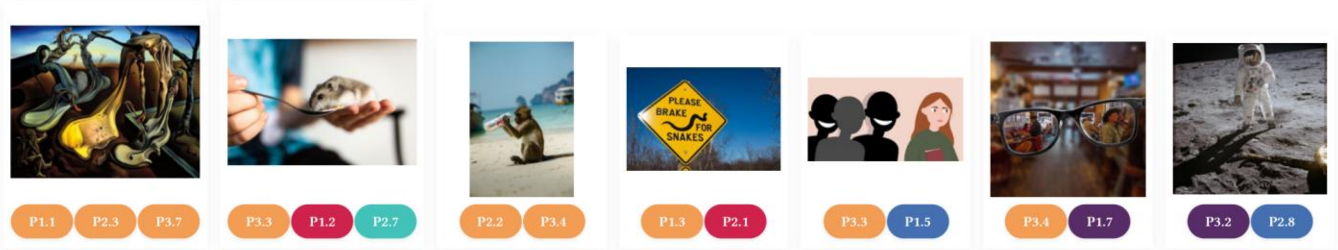
The interaction

Participants’ accounts were examined alongside the relation enacted in the chat logs. This pairing kept shifts in role, control, and responsibility tied to the exchange.

01B Relational Dissonance Finding and design direction



PARTICIPANT ELICITATION ARTIFACTS



SYNTHESIS

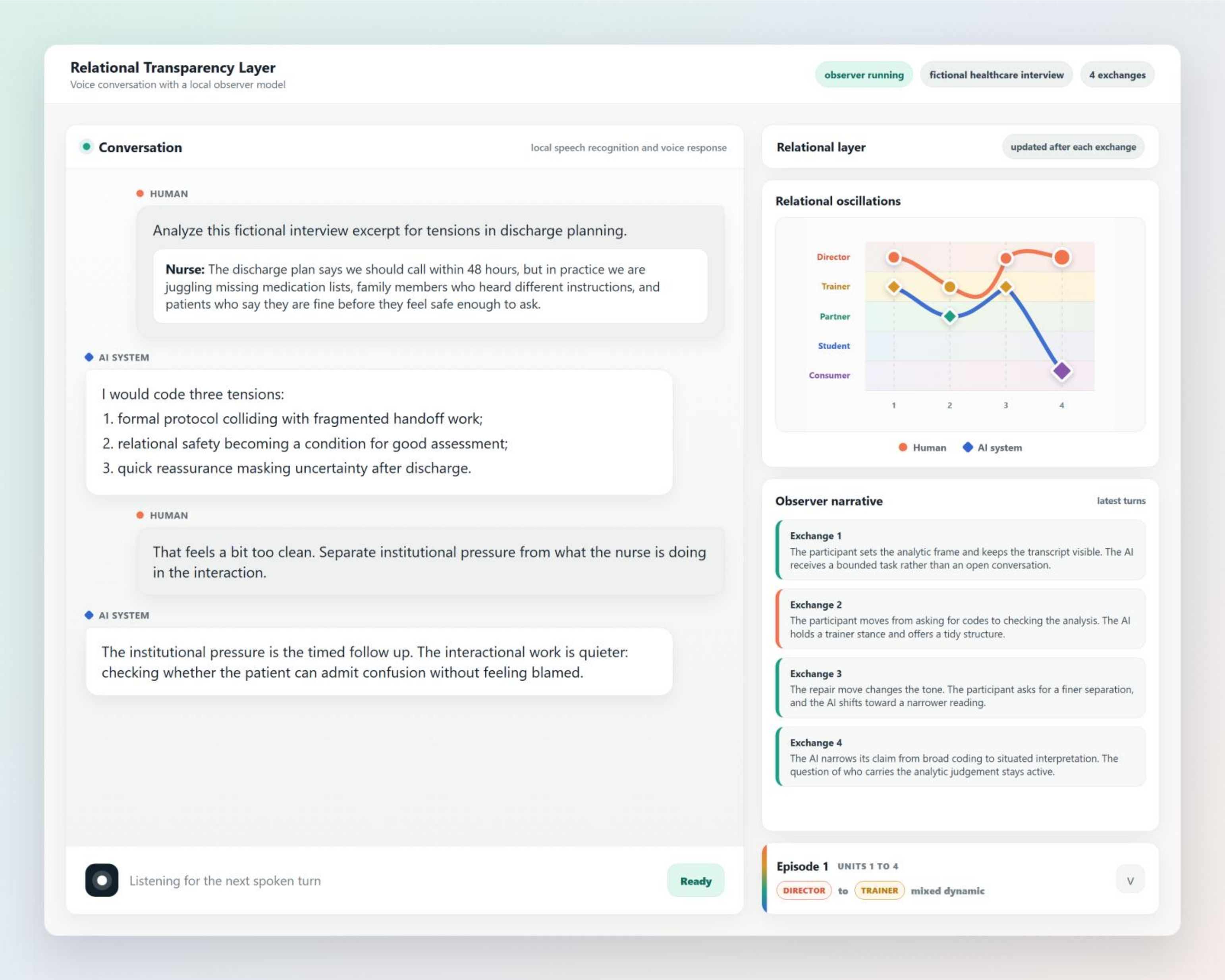
Our analysis found that participants' stated position toward AI often differed from the relation enacted in their chat logs.

Five recurring configurations located shifts in role, control, and responsibility in the interaction itself.

DESIGN RESPONSE

The findings informed the Relational Transparency project, where a separate observer account can be examined against the conversation.

Images in the lower strip were selected and discussed by participants during the workshop.



02 ACCEPTED NORDICHI 2026 DEMO

Relational Transparency Layer

A voice based desktop prototype for examining a provisional model account against the conversation and the user's own experience.

ROLE
Project lead and sole demo paper author.

SEPARATE
Conversation and observation run as separate processes.

TRACE
The account is linked to completed turns and visible interaction evidence.

QUESTION
A demo user can compare the observer account with the transcript and their own experience.

STATUS
Accepted demo, NordiCHI 2026. Formal participant evaluation remains future work.

WATCH THE 55 SECOND DEMO

1/4

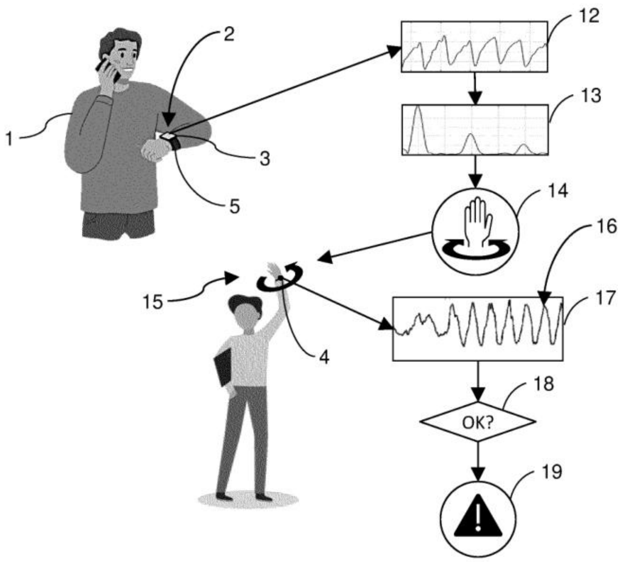


FIG. 1

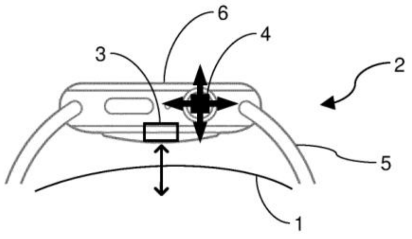


FIG. 2

WO2024235471A1

| guided interaction for wearable sensing accuracy

HUMAN MACHINE INTERACTION

HMI research and technology planning

HMI Technology Planning Specialist and HMI Research Intern at Huawei, 2022 to 2023.

RESEARCH PRACTICE

Researched emerging technologies across AI, smart rings, XR, haptics, and MEMS. Authored more than 30 technology reports and initiated projects with 11 plus research institutions and 6 companies.

PUBLIC RECORD

Named co inventor on five published international patent applications under the PCT.

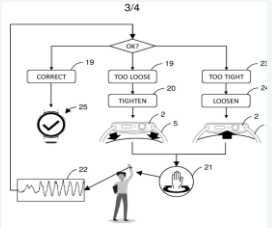
- WO2024235471A1 wearable sensing accuracy
- WO2024213244A1 distributed smart devices
- WO2024188442A1 multisensory smart insole
- WO2024179676A1 communication haptics
- WO2024179677A1 activity related haptics

PUBLIC EVIDENCE

Patent records confirm co inventorship and publication. Confidential methods and product outcomes remain outside this portfolio.

EMPLOYMENT RECORD SCALE (CV)

30+	5	11+	6
technology reports	PCT applications	research institutions	companies



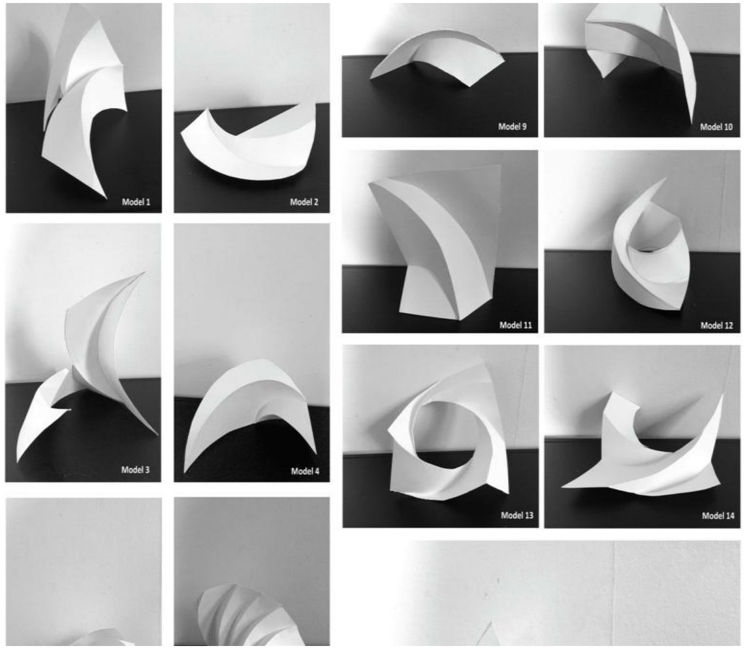
PUBLIC EXAMPLE

A closed feedback loop for wearable sensing

The published drawing connects device fit, sensing accuracy, corrective feedback, and another measurement cycle.

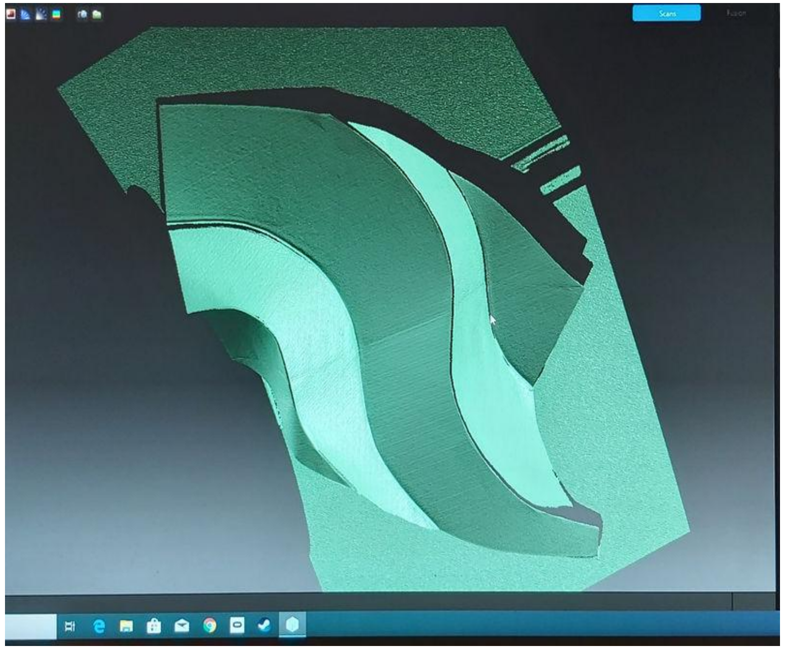
04 Curved Folding A physical to digital research sequence

A design process moved between paper models, scanning, simulation, and plywood fabrication.



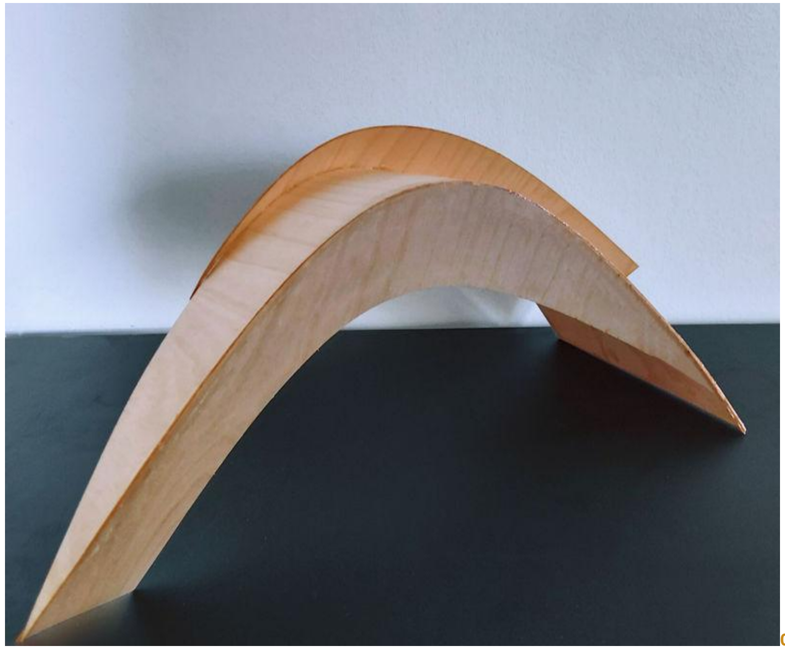
01

Explore and select



02

Translate and test



03

Fabricate and inspect

42	5	2	CHI 2021
paper models	shortlisted	plywood prototypes	verified publication

PLACE IN THE TRAJECTORY

This project established a feedback oriented research through design practice built around iterative translation and inspectable artifacts. Those commitments continue in the later human AI interaction research.

Role: first author; the paper states that the design process was mainly carried out by the first author.

Exploring a Feedback-Oriented Design Process Through Curved Folding | DOI 10.1145/3411764.3445639

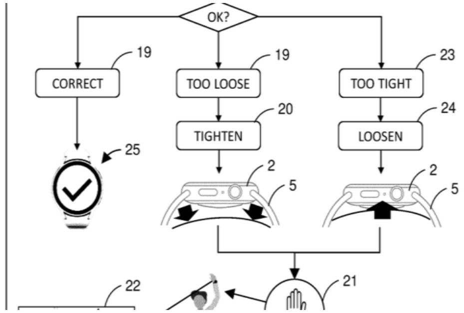
R

Research profile One trajectory across four research settings

RESEARCH THROUGH DESIGN

Feedback oriented inquiry

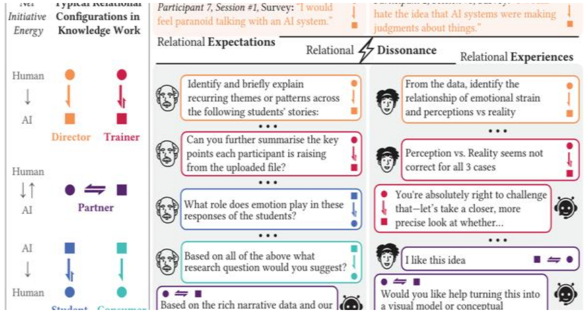
Iterative artifacts across physical and digital processes



INDUSTRY RESEARCH

HMI technology planning

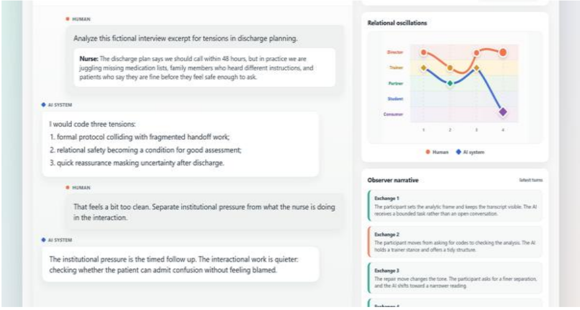
Technology reports, external collaboration, and 5 PCT applications



PARTICIPANT RESEARCH

Relational Dissonance

3 workshops, 22 researchers, and multiple qualitative evidence sources



PROTOTYPE INQUIRY

Relational Transparency

Accepted NordiCHI 2026 demo with evaluation pending

RESEARCH TRAJECTORY

Feedback oriented design inquiry, HMI technology planning, Relational Dissonance, and Relational Transparency form one trajectory. Across these settings, artifacts and interaction evidence support close study of how a process changes.

CURRENT RESEARCH PROGRAM

The broader research program focuses on human AI interaction.

Across the program, I use qualitative inquiry, interaction records, and research prototypes. Relational Dissonance and Relational Transparency examine shifts in role, control, interpretation, and responsibility across an exchange.

CURRENT ROLE
Postdoctoral Researcher, Responsible Use of AI, Aalto University

RESEARCH LEADERSHIP
Project lead, Relational Transparency Layer

FUNDING
Lead proposal author for 150,000 euro project; project lead for 16,000 euro project

CONCURRENT SERVICE
Data Agent, Aalto University Open Research Network

EMRECAN GULAY

Postdoctoral Researcher · Doctor of Arts · Espoo, Finland

Seeking permanent applied research, UX research, responsible AI, human AI interaction, or academic roles in Finland.

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Live research portfolio
emrecan713.github.io
LinkedIn
Google Scholar

METHODS ACROSS THE PORTFOLIO
Interviews, workshops, surveys, chat logs, participant artifacts, observations, interpretive analysis, research through design, and prototype development.

SETTINGS
Human AI interaction, responsible AI, qualitative knowledge work, HMI technology planning, physical and digital design processes, and research data practice.

VERIFIED WORK Relational Dissonance · CHI 2026 | Curved Folding · CHI 2021 | Relational Transparency Layer · accepted NordiCHI 2026 demo